$\Gamma \frac{1}{e^z}$.

Since $e^z = 1 + z + \frac{1}{2!}z^2 + \frac{1}{3!}z^3 + \cdots$,

$e^{\frac{1}{z}} = 1 + \frac{1}{z} + \frac{1}{2!\,z^2} + \cdots \quad (|z|>0)$

So, $\mathrm{Res}(e^{\frac{1}{z}}, 0) = 1$, therefore $\int_{|z|=1} e^{\frac{1}{z}}\, dz = 2\pi i$.

「Thm. Residue Theorem.

Let f be an analytic inside and on a simple closed contour C except iso. sing. $z_1, \cdots, z_n$.

Then, $\displaystyle\int_C f(z)\,dz = 2\pi i \cdot \sum_{k=1}^{n} \operatorname{Res}(f, z_k)$.

「Lemma. If f has a pole of order m at z_0, then

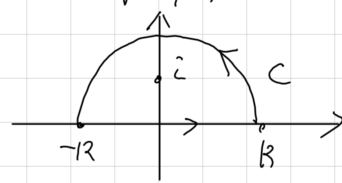
$$\operatorname{Res}(f, z_0) = \frac{1}{(m-1)!} \cdot \lim_{z \to z_0} \left((z-z_0)^m \cdot f(z) \right)^{(m-1)}$$

Example.

Evaluate $\displaystyle\int_{-\infty}^{\infty} \frac{1}{x^2+1}\,dx = \pi.$

Let $f(z) = \dfrac{1}{z^2+1}$. It has two isolated singular $z = \pm i$.

And, $\dfrac{1}{z^2+1} = \dfrac{1}{z-i}\cdot\dfrac{1}{z+i}$, so it has a simple pole, so



$\operatorname{Res}(f, i) = \lim_{z\to i}(z-i)\cdot\dfrac{1}{z^2+1} = \dfrac{1}{2i}$

and similarly $\operatorname{Res}(f,-i) = -\dfrac{1}{2i}$.

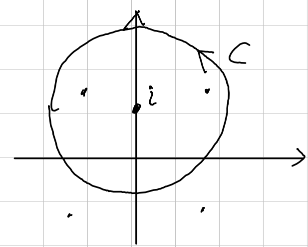
Now, for half-circle C containing i,

$$\int_C \frac{1}{z^2+1}\,dz = \int_{-R}^{R}\frac{1}{x^2+1}\,dx + \int_0^{\pi}\frac{iRe^{i\theta}}{R^2e^{2i\theta}+1}\,d\theta$$

And $$\left|\int_0^{\pi}\frac{iRe^{i\theta}}{R^2e^{2i\theta}+1}\,d\theta\right| \le \int_0^{\pi}\left|\frac{iRe^{i\theta}}{R^2e^{2i\theta}+1}\right|d\theta \le \int_0^{\pi}\frac{R}{R^2-1}\,d\theta = \frac{R\pi}{R^2-1} \to 0 \text{ as } R\to\infty.$$

$(|R^2e^{i\theta}+1| \ge |R^2e^{i\theta}|-1)$ so,

$$\int_{-\infty}^{\infty}\frac{1}{x^2+1}\,dx = \int_C\frac{1}{z^2+1}\,dz = 2\pi i\cdot\frac{1}{2i} = \pi.$$



Evaluate $\int_C \frac{1}{z^4+4}\,dz$ where $C=\{z\in\mathbb{C} \mid |z-i|=2\}$.

proof. Since z^4+4 has four roots

$$\sqrt{2}\cdot\exp\left(\frac{1}{4}\cdot\pi i\right),\ \sqrt{2}\exp\left(\frac{3}{4}\cdot\pi i\right),\ \sqrt{2}\exp\left(\frac{5}{4}\pi i\right),\ \sqrt{2}\exp\left(\frac{7}{4}\pi i\right),$$

so, by the Residue Theorem,

$$\mathrm{Res}(f,a_0)+\mathrm{Res}(f,a_1)=\lim_{z\to a_0}\frac{z-a_0}{z^4+4}+\lim_{z\to a_1}\frac{z-a_1}{z^4+4}$$

$$=\frac{1}{(a_0-a_1)(a_0-a_2)(a_0-a_3)}+\frac{1}{(a_1-a_0)(a_1-a_2)(a_1-a_3)}$$

Since $a_0=\sqrt{2}\left(\frac{1}{\sqrt{2}}+i\cdot\frac{1}{\sqrt{2}}\right)=1+i$

$a_1=\sqrt{2}\left(-\frac{1}{\sqrt{2}}+i\frac{1}{\sqrt{2}}\right)=-1+i$

$a_2=-1-i$

$a_3=1-i$,

$$\mathrm{Res}(f,a_0)+\mathrm{Res}(f,a_1)=\frac{1}{2\cdot(2+2i)(2i)}+\frac{1}{(-2)\cdot(2i)\cdot(-2+2i)}$$

So $\int_C \frac{1}{z^4+4}\,dz = 2\pi i\cdot(\mathrm{Res}(f,a_0)+\mathrm{Res}(f,a_1))$

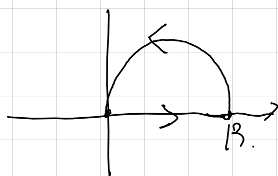
$$=\frac{\pi}{4+4i}+\frac{\pi}{4-4i}=\frac{8\pi}{32}=\frac{\pi}{4}.\ \blacksquare$$

「Evaluate $\int_0^\infty \frac{1}{1+x^6}\,dx$.

Let $f(z)=\frac{1}{1+z^6}$. Then, since $z^6=-1$ has six roots as

$$a_k = \exp\left(2\pi i\cdot\frac{k}{6}+\frac{\pi}{6}i\right) \quad (k=0,1,2,\cdots,5).$$

Let a Countor C_R as



As $R\to\infty$, C_R Contains only a_0, so by the Residue Theorem,

$$\int_{C_R} f(z)\,dz = 2\pi i\cdot \operatorname{Res}(f;a_0) = \lim_{z\to a_0}(z-a_0)\cdot f(z) = 1/\prod_{i=1}^{4}(a_0-a_i)$$

Meanwhile,

$$\int_{C_R} f(z)dz = \int_0^R \frac{1}{1+x^6}\,dx + \int_0^\pi \frac{iRe^{i\theta}}{1+(R\cdot e^{i\theta}+\frac{R}{2})^6}\,d\theta$$

$$z(t)=R\cdot e^{i\theta}+\frac{R}{2}$$

「Evaluate

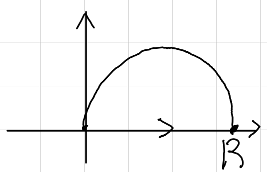
$$\int_0^\infty \frac{1}{1+x^n}\,dx \quad \text{for } n = 2, 3, \cdots$$

Proof. Let $f(z) = \dfrac{1}{1+z^n}$.

Since $z^n = -1$ has roots as

$$a_k = \exp\left(\frac{1}{n}\cdot\left(2k\pi i + \pi i\right)\right) \quad (k = 0, 1, \cdots n)$$

Construct a Counter C_R as



$$0 \le \frac{\pi \cancel{i}}{n}\cdot(2k+1) \le \frac{\pi}{2}\cancel{i}$$

$$0 \le 4k+2 \le n$$

As $R \to \infty$, the Counter Contains inside

$$a_1, a_2, \cdots, a_m, \quad \text{where } m = \max\{i \in \{0, \cdots, n-1\} \mid 4i+2 \le n\}.$$

Now, by the Residue Theorem,

$$\int_{C_R} \frac{1}{1+z^n}\,dz = 2\pi i \cdot \Sigma$$